

Kubernetes etcd Disk Full / Quota Exceeded Runbook

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Category	kubernetes	Severity	P1 - Critical
Domain	Container Orchestration	Est. Duration	30-60 minutes
Tags	kubernetes, etcd, disk, quota, defrag, compaction, control-plane		

Overview

etcd enforces a storage quota (default 2GB). When the quota is exceeded, etcd enters a read-only alarm state, rejecting all write operations. This makes the Kubernetes API server read-only — no new pods, deployments, or config changes can be applied. Immediate action required.

Prerequisites

- SSH access to etcd control plane nodes
- etcdctl installed and ETCDCTL_API=3 set
- etcd TLS certificates accessible
- Cluster maintenance window (API writes will be down during compaction)

Response Steps

Step 1: Confirm etcd Alarm State

Check if etcd has triggered the NOSPACE alarm that causes read-only mode.

```
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key alarm list

ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key endpoint status
```

NOTE: ALARM: NOSPACE confirms the quota is exceeded.

Step 2: Check etcd Disk Usage

Verify the actual disk usage and available space on the etcd data directory.

```
df -h /var/lib/etcd
du -sh /var/lib/etcd/member/
ls -lh /var/lib/etcd/member/snap/
```

Step 3: Compact etcd Revision History

Compact old revision history. etcd keeps all historical revisions — compaction removes all revisions below the specified one.

```
# Get current revision:
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key endpoint status --write-out=json | python3 -c
"import sys,json; print(json.load(sys.stdin)[0]['Status']['header']['revision'])"
```

```
# Compact to current revision (replace ):
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key compact
```

NOTE: Compaction is safe — it only removes old revision data, not current state.

Step 4: Defragment etcd

After compaction, defragment to reclaim the physical disk space. Do each etcd member one at a time.

```
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key defrag
df -h /var/lib/etcd
```

NOTE: Defrag takes etcd offline briefly for that member. Do NOT defrag all members simultaneously.

Step 5: Disarm the NOSPACE Alarm

After freeing space, disarm the alarm to restore write access.

```
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key alarm disarm
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key alarm list
```

NOTE: SUCCESS: alarm list returns empty. kubectl commands now work normally.

Step 6: Increase etcd Quota Limit

Increase the etcd quota to prevent recurrence. Edit the etcd static pod manifest.

```
# Edit /etc/kubernetes/manifests/etcd.yaml and add/update:
# - --quota-backend-bytes=8589934592 (8GB)
vi /etc/kubernetes/manifests/etcd.yaml
# etcd will restart automatically
```

NOTE: Recommended: set to 4-8GB depending on cluster size. Maximum supported is 8GB.

Rollback Steps

Rollback Step 1: Restore etcd from Snapshot

If compaction/defrag corrupted the database, restore from the most recent snapshot.

```
ETCDCTL_API=3 etcdctl snapshot restore /path/to/snapshot.db --data-dir=/var/lib/etcd-restore
# Replace /var/lib/etcd with restored data
systemctl restart etcd
```

Rollback Step 2: Revert Quota Change

If the quota increase causes instability, revert to the default 2GB.

```
vi /etc/kubernetes/manifests/etcd.yaml
# Set --quota-backend-bytes=2147483648
```

Step Dependencies

Step	Depends On	Can Parallelize With
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Step 1: Confirm Alarm State	—	—
Step 2: Check Disk Usage	Step 1	—
Step 3: Compact Revision History	Step 1, Step 2	—
Step 4: Defragment etcd	Step 3	—
Step 5: Disarm NOSPACe Alarm	Step 4	—
Step 6: Increase Quota Limit	Step 5	—

Maintained by Platform Engineering. For updates ping #platform-oncall.